

Report of Project 4: Non-premixed Laminar Flames

Russ, Rachel

*Department of Aeronautics and Astronautics
Purdue University
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1 Counter Jet Flow of Hydrogen-Air Mixture

1.1 Effect of Different Transport Models – Q1(a)

The different transport models cause FLUENT to solve the cases slightly differently which causes the results to differ. The full multicomponent diffusion changes how the species diffusion fluxes are computed with the Maxwell-Stefan equation which can change the species profiles. It is normally used for complex mixtures. The thermal diffusion makes it so FLUENT includes the Soret Effect which allows the temperature gradient to affect the distribution of particles as well as the regular concentration gradients and turbulent mixing. It is normally used when there are large temperature gradients, light species, or compressible flows.

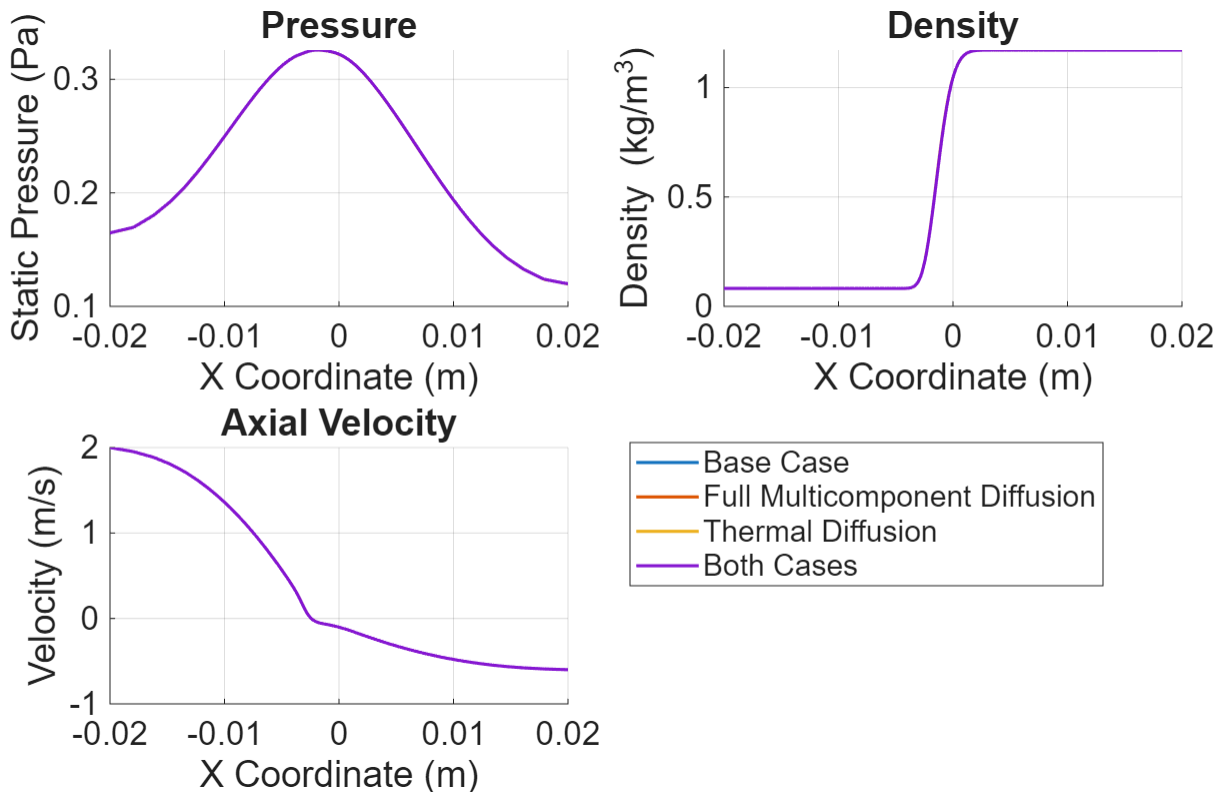


Figure 1 Difference of Pressure, Density, and Axial Velocity with Different Transport Models

Fig. 1 shows the pressure, density, and axial velocity differences in each case included a base case and when both of the cases are accounted for. Because this model is a cold flow at 300K, the reactions do not occur which means that these models are not very different. This is because the things that the different diffusion models account for are negligible.

Table 1 Percent Difference of Different Outputs for Different Transport Models

Value	Pressure	Density	Axial Velocity
Base-Multi	0.0095	0.7280	0.6018
Base-Therm	2.9084e-05	0.0011	3.0726e-04
Base-Both	0.0095	0.7280	0.6018
Multi-Therm	0.0376	0.4134	0.1868
Multi-Both	7.5859e-08	4.1516e-05	1.6819e-05
Therm-Both	0.0095	0.7270	0.6027

Table 1 shows the percent differences between certain models for each pressure, density, and axial velocity. These percent differences are very low, under 1 percent, which shows that these models do not differ much. This helps support the claim that the base case works just as well with less computing power as the full multicomponent diffusion and thermal diffusion.

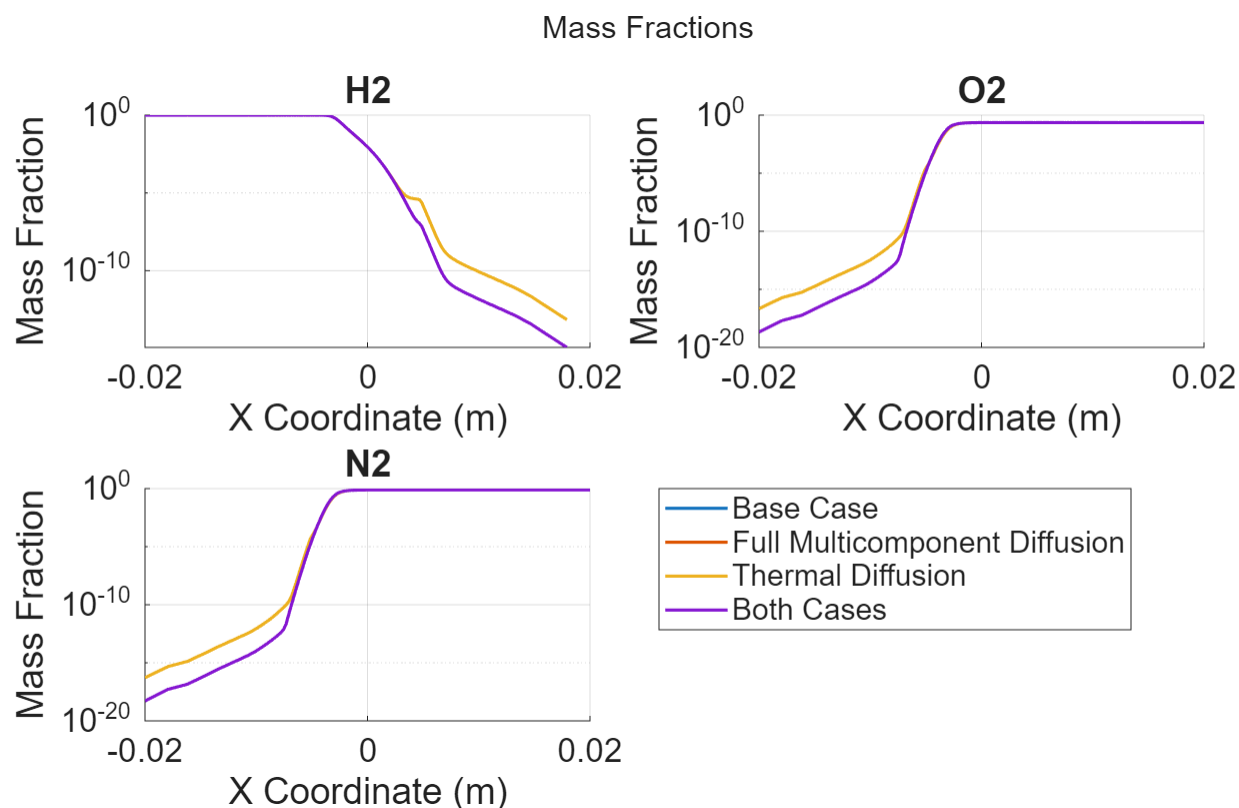


Figure 2 Difference of Mass Fractions with Different Transport Models

Fig. 2 shows the mass fractions of H_2 , O_2 , and N_2 . Because this case is nonreacting due to the low temperature of 300K, there are products that are not present such as H_2O , HO_2 , H , O , OH ,

and H_2O_2 . As shown by the figure, all the cases match well unless the mass fractions are very low which is shown by where the thermal diffusion model differs from the other cases. However, this is when the mass fractions are less than 10^{-10} for nitrogen and oxygen and 10^{-5} for hydrogen which means the change is negligible.

Table 2 Percent Difference of Mass Fractions with Different Transport Models

Species	H ₂	O ₂	N ₂
Base-Multi	0.5167	1.0511	67.8901
Base-Therm	1.4502e-06	0.0010	0.0010
Base-Both	0.5168	1.0511	67.8901
Multi-Therm	1.6317	14.4654	1.8248e-05
Multi-Both	1.8295e-05	1.8248e-05	1.8248e-05
Therm-Both	0.5172	1.0509	67.9000

Table 2 shows the percent differences between the different mass fractions and the different transport models. The percent difference for hydrogen is low, under 2%, for each of the different model comparisons. For oxygen, there is a larger percent difference with the comparison of the full multicomponent diffusion and the thermal diffusion case. For nitrogen, there are large percent differences with the cases where both models are accounted for. This is due to the presence of hydrogen.

1.2 Effect of Different Discretization Schemes – Q1(b)

The different discretization schemes analyzed are first order upwind, 2nd order upwind, and QUICK (Quadratic Upstream Interpolation for Convective Kinematics). First order upwind uses upstream values to compute fluxes, and it works well for courser meshes and it is highly stable. However, it is not as accurate as other schemes, and it does not work well for sharp gradients. Second order upwind uses linear interpretation from neighboring cells, and it is the default setting in FLUENT. It is higher accuracy, but it is less stable than first order. QUICK uses quadratic interpolation, and it is upwind based like the other schemes. It works well for smooth flows, but it is not as accurate for sharp gradients, and it is more expensive.

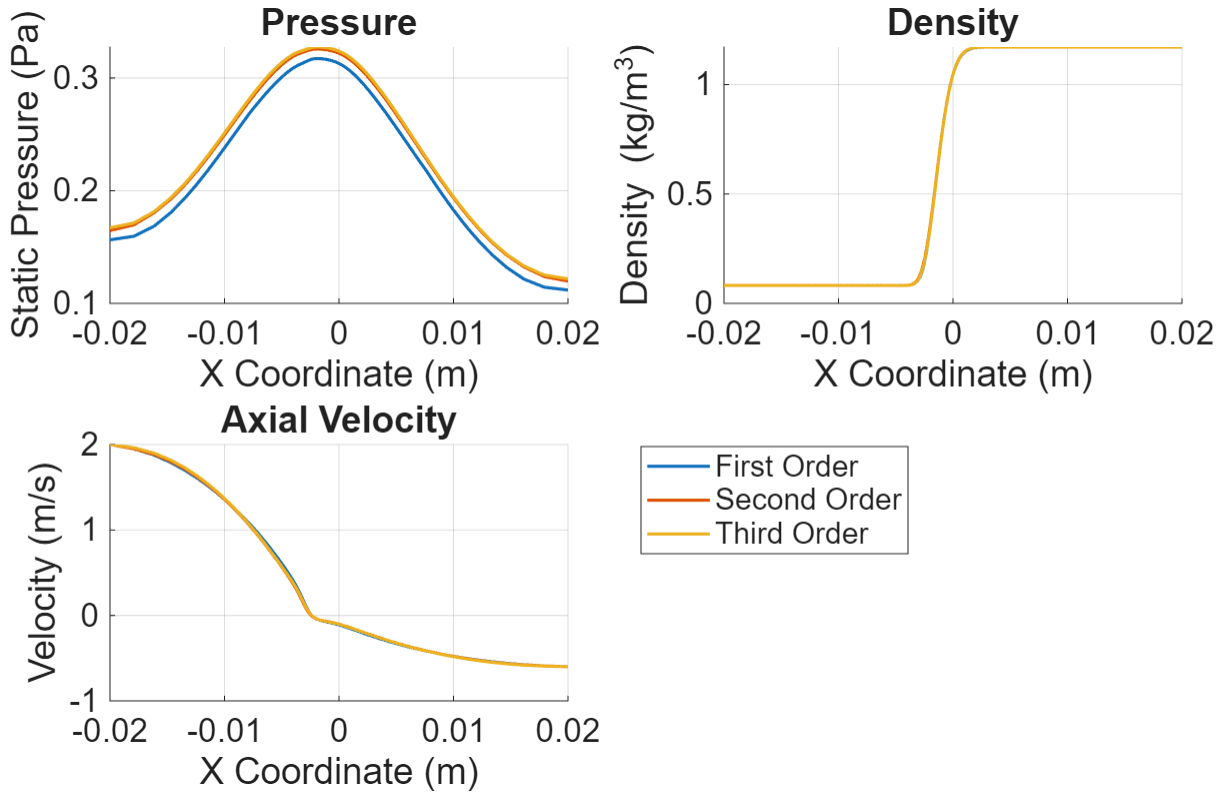


Figure 3 Difference of Pressure, Density, and Axial Velocity with Different Discretization Schemes

Fig. 3 shows the differences in pressure, density, and axial velocity for different discretization schemes. The second order and third order models show higher pressure values than the first order model. The density for all three models is very similar, and the axial velocity values are also very similar; however, near the x coordinate -0.01m, the difference schemes differ slightly more.

Table 3 Percent Difference of Different Outputs with Different Discretization Schemes

Value	Pressure	Density	Axial Velocity
1 st – 2 nd	6.8682	0.1693	7.8411
1 st – 3 rd	8.5296	0.5923	5.2346
2 nd – 3 rd	0.5582	2.9895	0.5485

Table 3 shows the percent difference between each scheme for different outputs. For pressure, there is a larger percent difference between the first and second/third order schemes and a smaller difference between the second and third order schemes. For density, the percent differences are low with the difference between second and third order being larger. For axial velocity, there is the same trend as with pressure. This shows that there is more of a difference between the first and second order schemes as there are the second and third order schemes, which explains why the default selection is the second order scheme as it provides accurate results without it being as expensive.

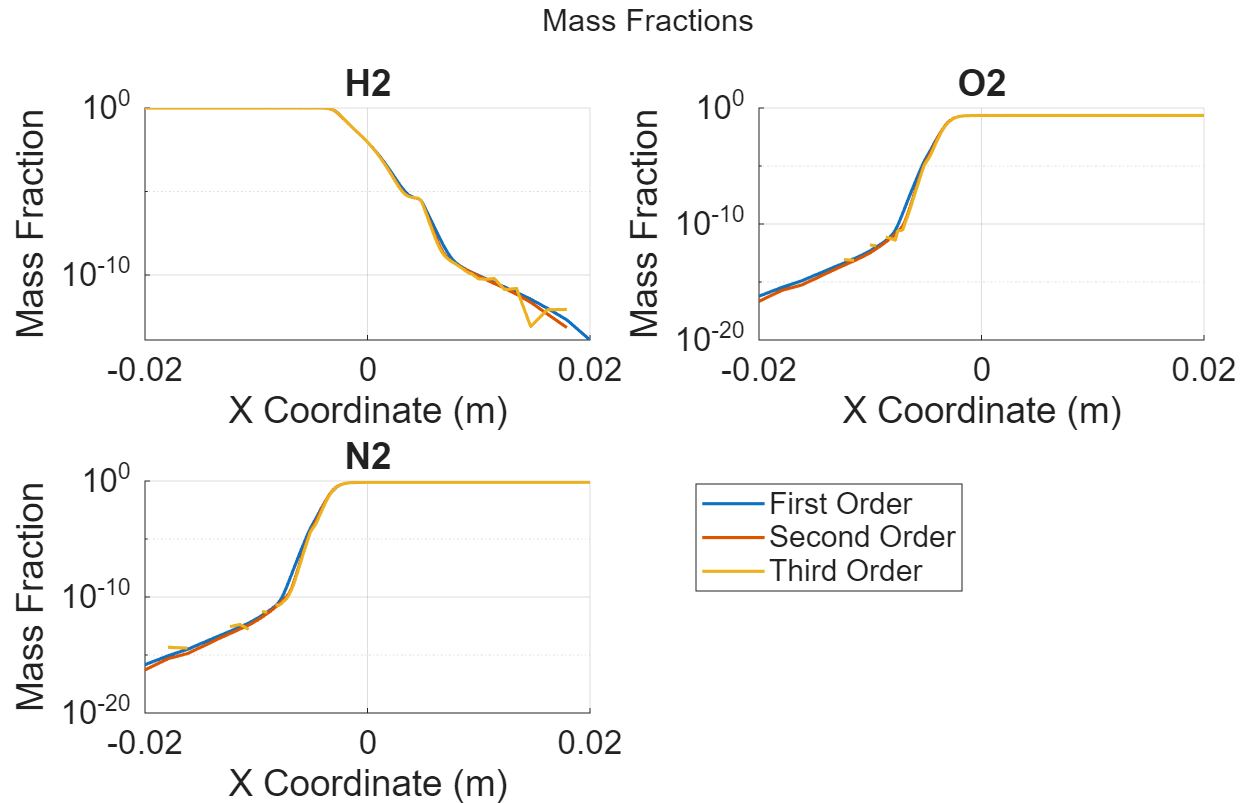


Figure 4 Difference of Mass Fractions with Different Discretization Schemes

Fig. 4 shows the mass fractions of H_2 , O_2 , and N_2 . Because this case is nonreacting due to the low temperature of 300K, there are products that are not present such as H_2O , HO_2 , H , O , OH , and H_2O_2 . The different schemes give very similar results for mass fractions greater than 10^{-10} . However, the differences are much greater at smaller mass fractions, especially the third order scheme. However, this happens at extremely low mass fractions, so the effects are negligible.

Table 4 Percent Difference of Mass Fractions with Different Discretization Schemes

Species	H_2	O_2	N_2
1 st – 2 nd	7.0068	12.9334	14.0246
2 nd – 3 rd	0.6558	6.7063	6.5113
1 st – 3 rd	2.9419	13.8943	14.6055

Table 4 shows the percent differences in mass fractions between the three non-zero reactants. The percent differences are highest between first and second order and are lowest between second and third order, which again shows why FLUENT has second order as the default. The largest percent errors are with oxygen and nitrogen which are also similar. This is because of the similar weights between the two.

1.3 Difference in Grid Size – Q1(c)

The grid is a major component on the accuracy of the results of the model. A finer grid produces more accurate results; however, it is much more expensive to run, which is why it is important to find a median between accuracy and cost with the grid. Table 5 shows the number of cells and faces for each of the three grids that are run. It is evident that the amount of cells gets exponentially larger with a finer grid which causes expense to also grow exponentially.

Table 5 Number of Cells and Faces of the Different Grids

	Coarsening (Courser Grid)	Original (Original Grid)	Refinement (Finer Grid)
Cells	1,100	4,400	17,600
Faces	2,320	9,040	35,680

Fig. 5 shows the differences in pressure, density, and axial velocity with different sizes of grids. In this non-reacting case, it is evedent that these changes are small. The biggest difference is pressure with the base grid having a lower pressure than the finer grid.

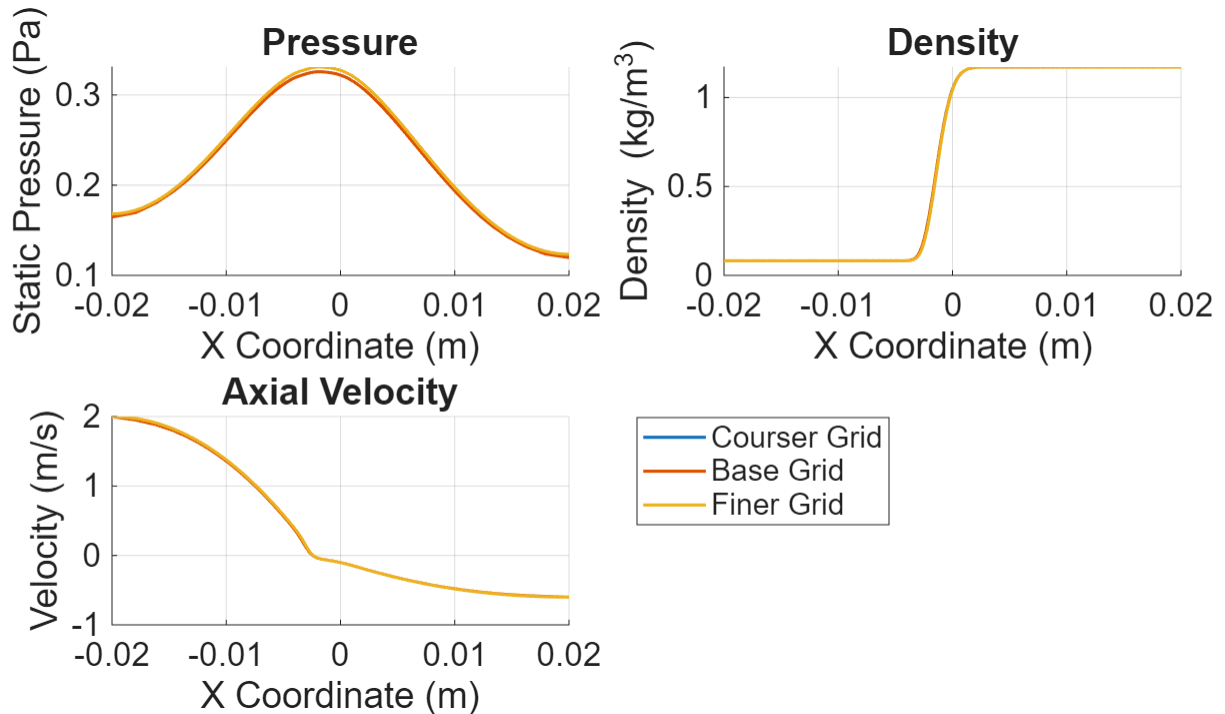


Figure 5 Difference of Pressure, Density, and Axial Velocity with Different Grid Sizes

However, density and axial velocity values do not vary much between the different grid sizes. Because of this, a course grid can be used for the non-reacting case with limited detriment to accuracy which is ideal. Table 6 shows the percent differences of pressure, density, and axial velocity with the different grid sizes.

Table 6 Percent Difference of Different Outputs with Different Grid Sizes

Value	Pressure	Density	Axial Velocity
Fine - Base	1.6549	0.0058	1.2891
Base - Course	0.0043	0.0041	0.2360
Fine - Course	1.6570	7.3529e-04	1.2897

It is evident that the differences in grid size do not make too much of a difference in the non-reacting case as the percent differences are lower than 2%. This is since there are no sudden changes in the non-reacting case, which is what a finer grid is good for.

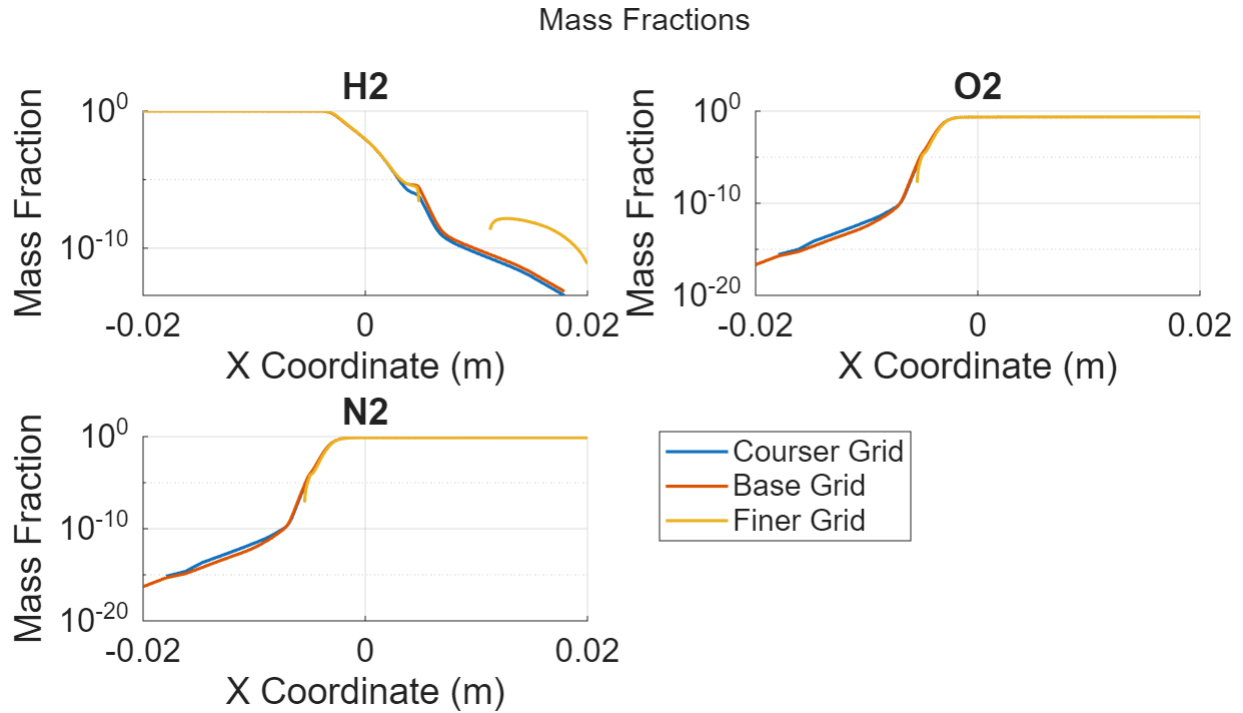


Fig. 6 shows the mass fractions of H_2 , O_2 , and N_2 . Because this case is nonreacting due to the low temperature of 300K, there are products that are not present such as H_2O , HO_2 , H , O , OH , and H_2O_2 . All the grids match up at mass fractions greater than 10^{-6} which is when the species is of importance. Any mass fractions lower than that the grids start to diverge slightly with the finer grid having wildly different results for H_2 , and it does not calculate mass fractions for O_2 and N_2 less than 10^{-6} .

Table 7 Percent Differences in Mass Fractions with Different Grid Sizes

Value	H_2	O_2	N_2
Fine – Base	5.8363	2.3913e-04	2.4045e-04
Base - Course	0.1997	16.8490	7.2375
Fine - Course	5.8363	5.5000e-05	5.6169e-05

Table 7 shows the percent differences in mass fractions for H₂, O₂, and N₂. For H₂, the smallest percent difference is from the base to course case. For O₂, the largest percent difference is the opposite with the base to course case. This is the same trend for N₂ as well. These differences are due to the major difference in weight between hydrogen and oxygen and nitrogen.

1.4 Scalar Dissipation Rate – Q1(d)

Scalar dissipation rate (χ) is the departure from chemical equilibrium and can be calculated by Eq.(1) below where $D = 2.5 \times 10^{-5} \text{ m}^2/\text{s}$, x is the x-coordinate, and ξ is the mixture fraction.

$$\chi \equiv 2D \left(\frac{d\xi}{dx} \right)^2 \quad (1)$$

There is a model for the scalar dissipation rate as well, shown in Eq.(2), where a is the strain rate.

$$\chi = \frac{a}{\pi} \exp\{-2[\text{erfc}^{-1}(2\xi)]^2\} \quad (2)$$

$$a = \frac{v_{fuel} + v_{ox}}{L} \quad (3)$$

The strain rate can be calculated in Eq.(3) where L is the gap distance (4 cm), v_{fuel} is 2 m/s, and v_{ox} is 0.6 m/s. The mixture fraction can be calculated by Eq.(4) where Y is the mass fraction of the species specified at the specified location.

$$\xi = \frac{Y_{H_2} - Y_{H_2,ox}}{Y_{H_2,fuel} - Y_{H_2,ox}} \quad (4)$$

Because there is no hydrogen in the oxidizer inlet, the mixture fraction equation simplifies to Eq.(5).

$$\xi = Y_{H_2} \quad (5)$$

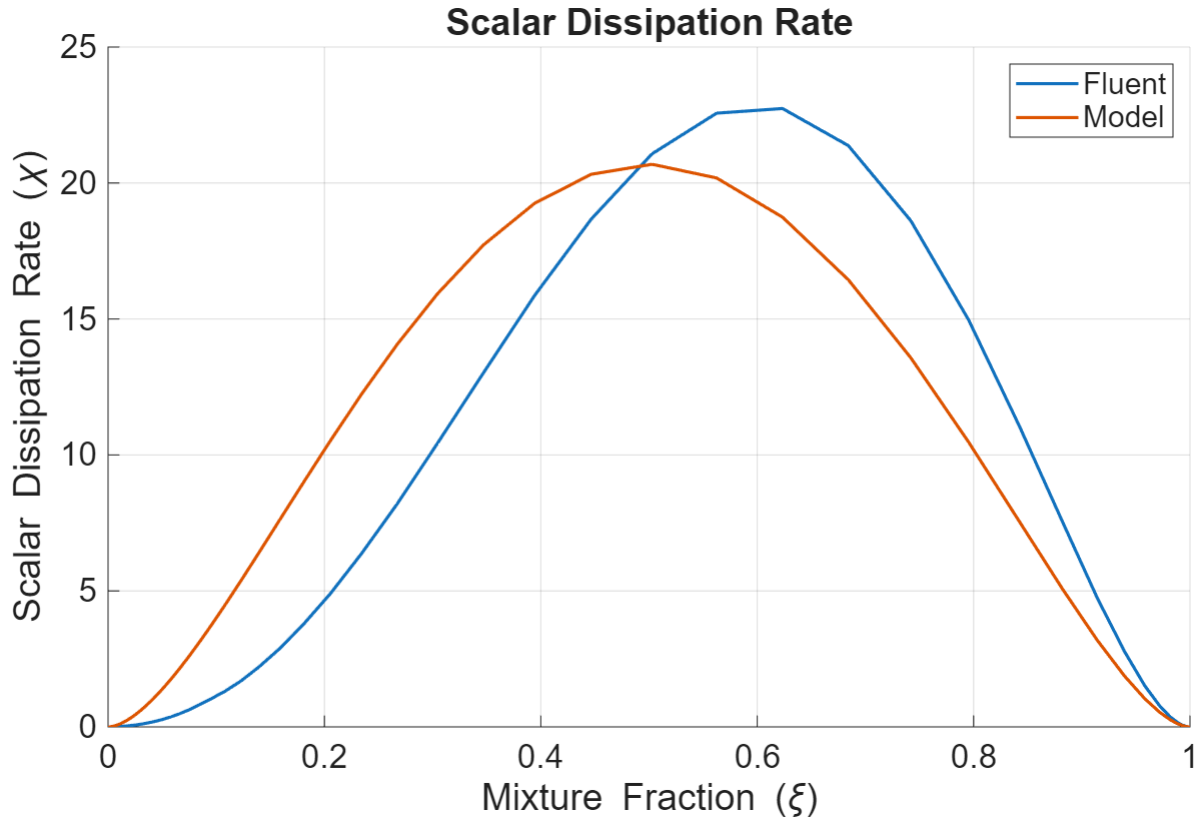


Figure 7 Scalar Dissipation Rate versus Mixture Fraction

Fig. 7 shows the scalar dissipation rate vs the mixture fraction. The models match up in shape; however, they are offset slightly, and the fluent model shows a higher scalar dissipation rate. The maximum scalar dissipation rate from FLUENT is 22.7392 1/sec at a mixture fraction of 0.6234. The maximum scalar dissipation rate from the model is 20.6891 1/sec at a mixture fraction of 0.50279. These values are similar, but the mixture fraction where the maximum occurs is more significantly different.

2 Hydrogen-Air Flame

2.1 Output Values Study - Q2(a)

Now the temperature is much higher, so the mixture ignites. This changes the results quite a bit compared to the cold flow case in section 1. Eq.(5) still holds as the equation for the mixture fraction due to the fact the fuel and oxidizer inlets are still separated. Fig. 8 shows the density, temperature, and axial velocity as functions of the physical space, or the x-coordinate system.

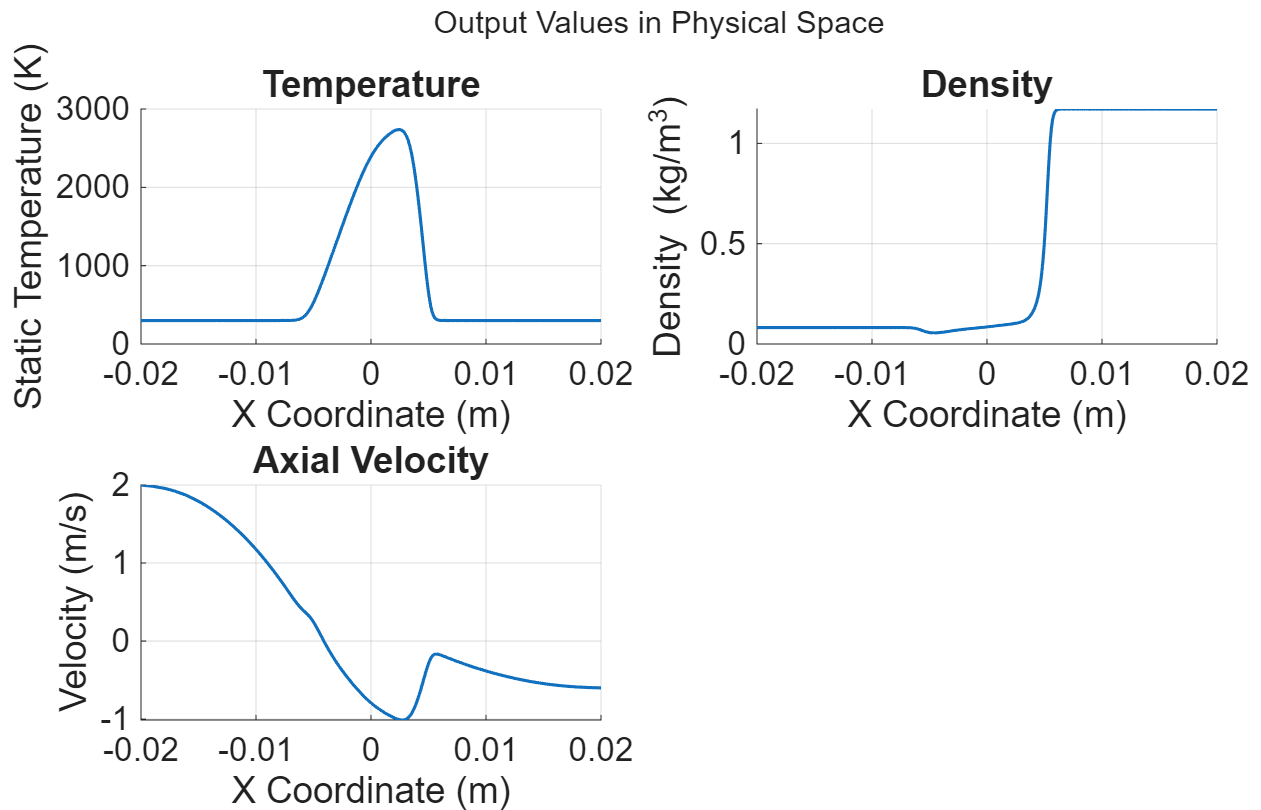


Figure 8 Temperature, Density, and Axial Velocity in the Physical Space

The flame front is clearly shown centered slightly greater than 0 m as shown by the temperature graph. The density is much greater near the oxidizer inlet as the oxidizer has a much greater density compared to hydrogen. The mixing is also shown by the axial velocity graph where the flame occurs.

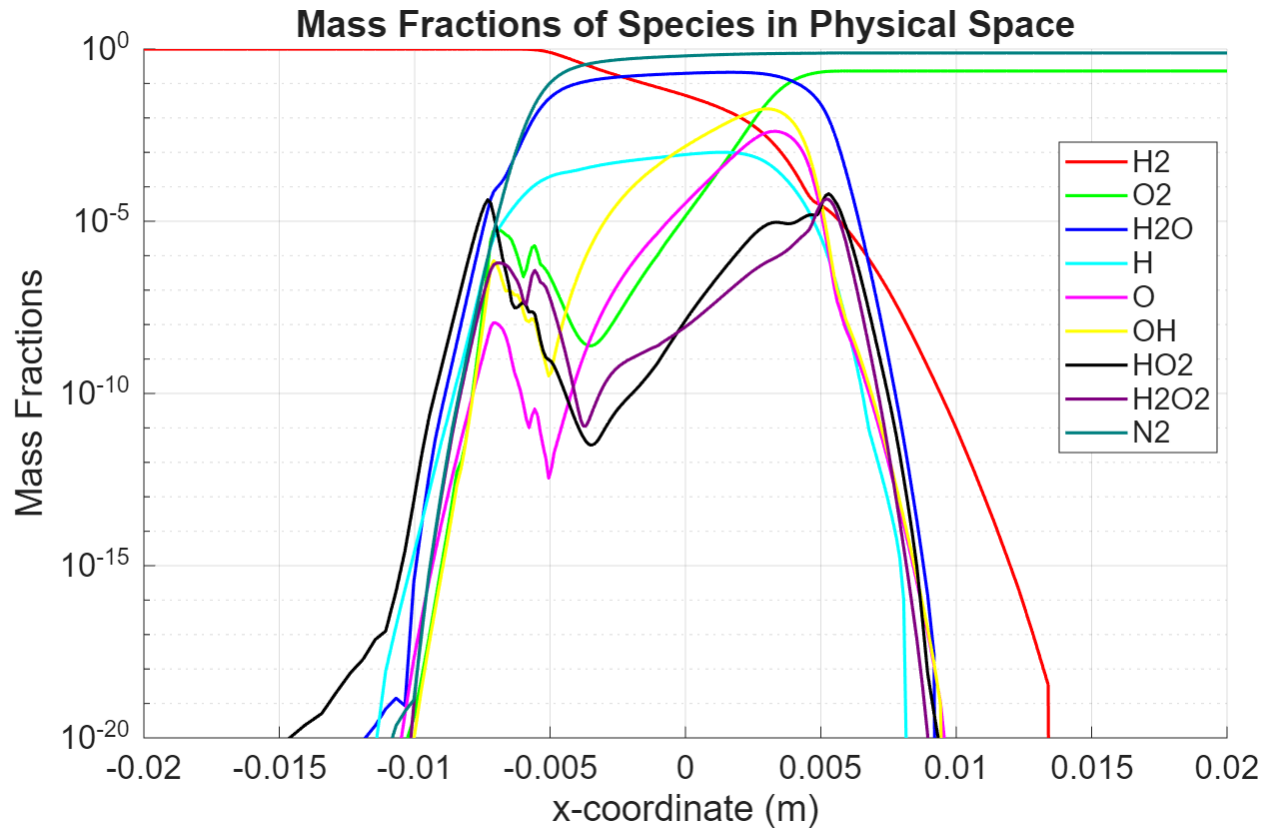


Figure 9 Mass Fractions of Each Species in the Physical Space

Fig. 9 shows the mass fractions of each species in the physical space. The oxidizers (O_2 and N_2) have very high mass fractions, on the order of one, near the right-hand side as this is where the inlet is. The same for the fuel (H_2) can be said on the left-hand side as this is also where its inlet is. The major products of the reactions (H_2O and H) are concentrated near where the flame occurs which is centered around an x-coordinate of 0 m. These major products have relatively larger mass fractions, greater than 10^{-4} , during the entire flame front which is expected. The radical products (O , OH , HO_2 , and H_2O_2) are concentrated at the beginning and end of the flame fronts but are smaller quantities than the major products. They are not as concentrated during the flame front which is to be expected as they get reabsorbed into the major products as they are unstable.

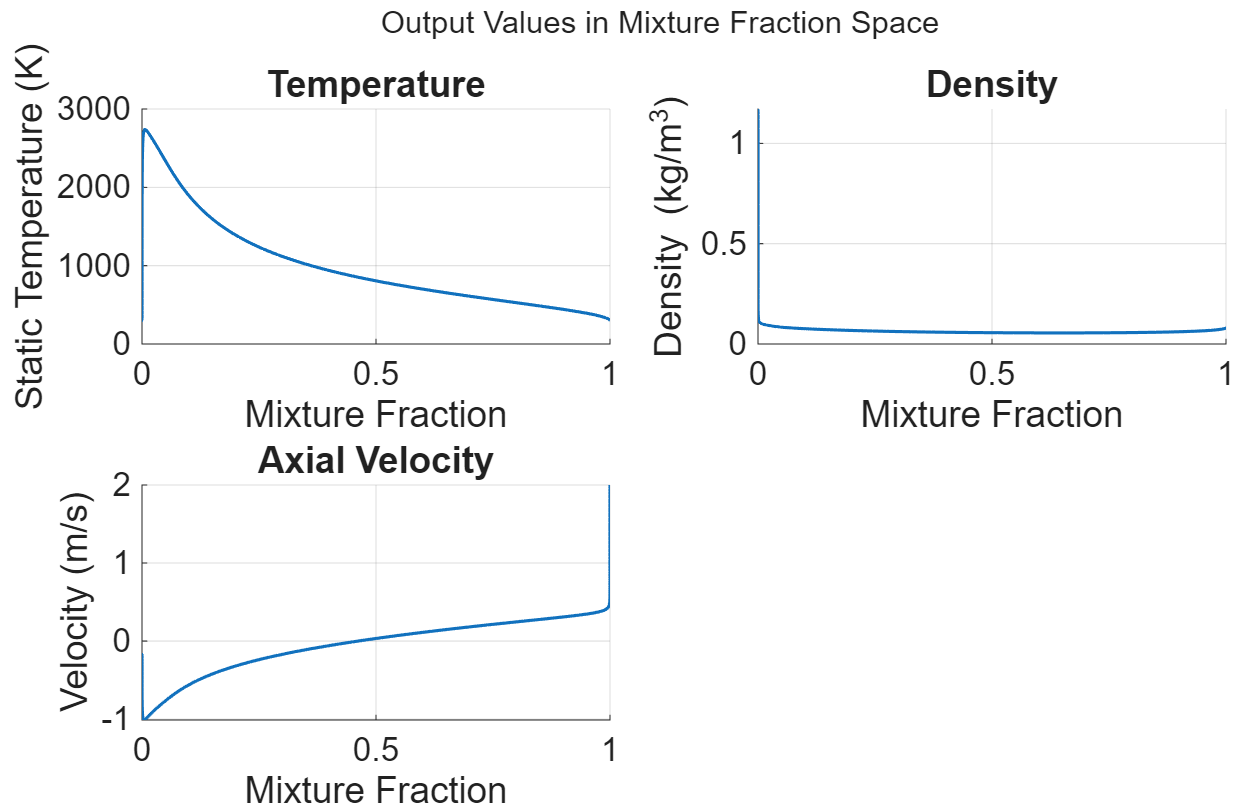


Figure 10 Temperature, Density, and Axial Velocity in the Mixture Fraction Space

Fig. 10 shows the temperature, density, and axial velocity of the reaction in the mixture fraction space. A mixture fraction of one is pure fuel (H_2) and a mixture fraction of zero is pure oxidizer (N_2 and O_2). The temperature is greater at a mixture fraction of almost zero (almost pure oxidizer) due to the fact the stoichiometric mixture fraction is very low as the result of the weight of hydrogen, which means it requires a lot of oxidizer. This relationship also holds for density as the greatest density is at a very low mass fraction due to the same reasons. The axial velocity is greatest at a high mixture fraction since the velocity of the fuel is higher to get the flame in the middle. The steady increase of velocity as the mixture fraction increases makes sense if the velocity of the oxidizer and fuel as well as their weights are considered.

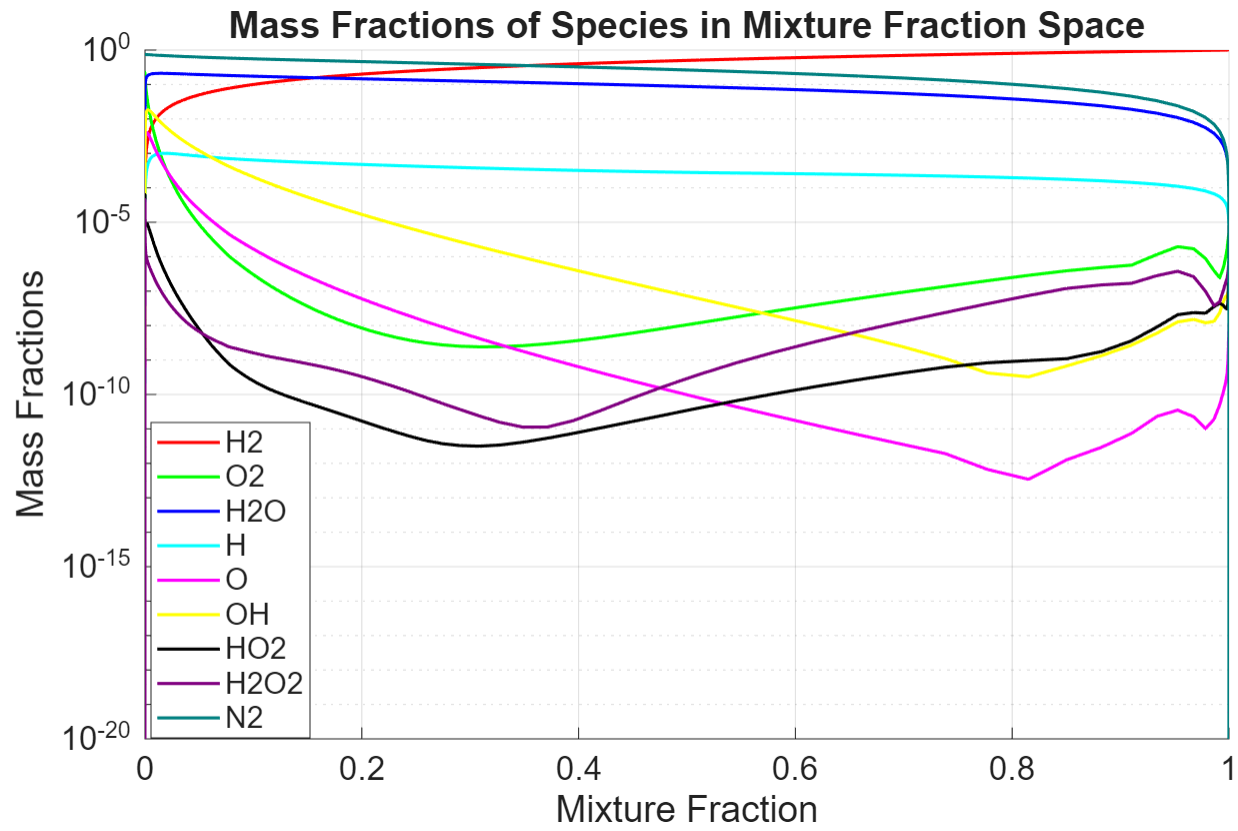


Figure 11 Mass Fractions of Each Speies in the Mixture Fraction Space

Fig. 11 shows the mass fractions of each of the species in the mixture fraction space. The oxidizers are more present when mixture fraction is small which makes sense as this is when more oxidizer (air) is present. The inverse is true for the fuel (H_2). The major products (H and H_2O) are present in quantities that align well with nitrogen because nitrogen is not consumed during the reaction. O and OH track each other well at an offset, and HO_2 , H_2O_2 , and O_2 also are offset similarly since the oxygen containing species are tightly coupled. O and OH are chain carrying radicals, so their involvement is not as linked to oxygen availability and more linked to temperature.

Table 8 Maximum Values and Where Occurs

	Temperature	Mass Fraction of H	Mass Fraction of H_2O
Maximum Value	2737.53	0.001004	0.211163
Mixture Fraction	0.0066282	0.018068	0.014710
Physical Location	0.00242	0.001390	0.002641

Table 8 shows the maximum temperature and mass fractions of H and H_2O as well as where these maximums occur in the mixture fraction space and the physical location. The maximum temperature and mass fraction is where the flame occurs, and it is at a very low mixture fraction.

2.2 Strain Rate Study – Q2(b)

The strain rate is the strain of the fluid per unit time. It is used to show how much the fluid is stretched with a high value meaning the fluid is very stretched. Eq.(6) shows the equation for strain rate (a) where v_{ox} is the velocity of the oxidizer (air), and v_{fuel} is the velocity of the fuel (H_2).

$$a = \frac{v_{fuel} + v_{ox}}{L} \quad (6)$$

Gap distance (L) is kept constant at 0.04 m. The ratio of fuel velocity to oxidizer (air) velocity is also kept constant to keep the flame in the middle of the window. It is shown in Eq.(7), which is calculated from the previous 2 m/s of fuel and 0.6 m/s of air.

$$\frac{v_{fuel}}{v_{ox}} = \frac{1}{0.3} \quad (7)$$

With both Eq.(6) and Eq.(7), both fuel and air velocities can be explicitly calculated if target strain rate is chosen. Table 9 shows the strain rate for each fuel and air velocities. These fuel and air velocities are inputs into FLUENT to simulate the different strain rates.

Table 9 Fuel and Air Velocity for Each Strain Rate

Strain Rate (a) (1/sec)	Fuel Velocity (v_{fuel}) (m/s)	Air Velocity (v_{ox}) (m/s)
1	0.0308	0.00924
10	0.30769	0.09231
50	1.53846	0.46154
100	3.0769	0.92307
500	15.3846	4.61538
1000	30.7692	9.23076
1500	46.1539	13.84615
2000	61.5385	18.46154
3000	92.3077	27.6923

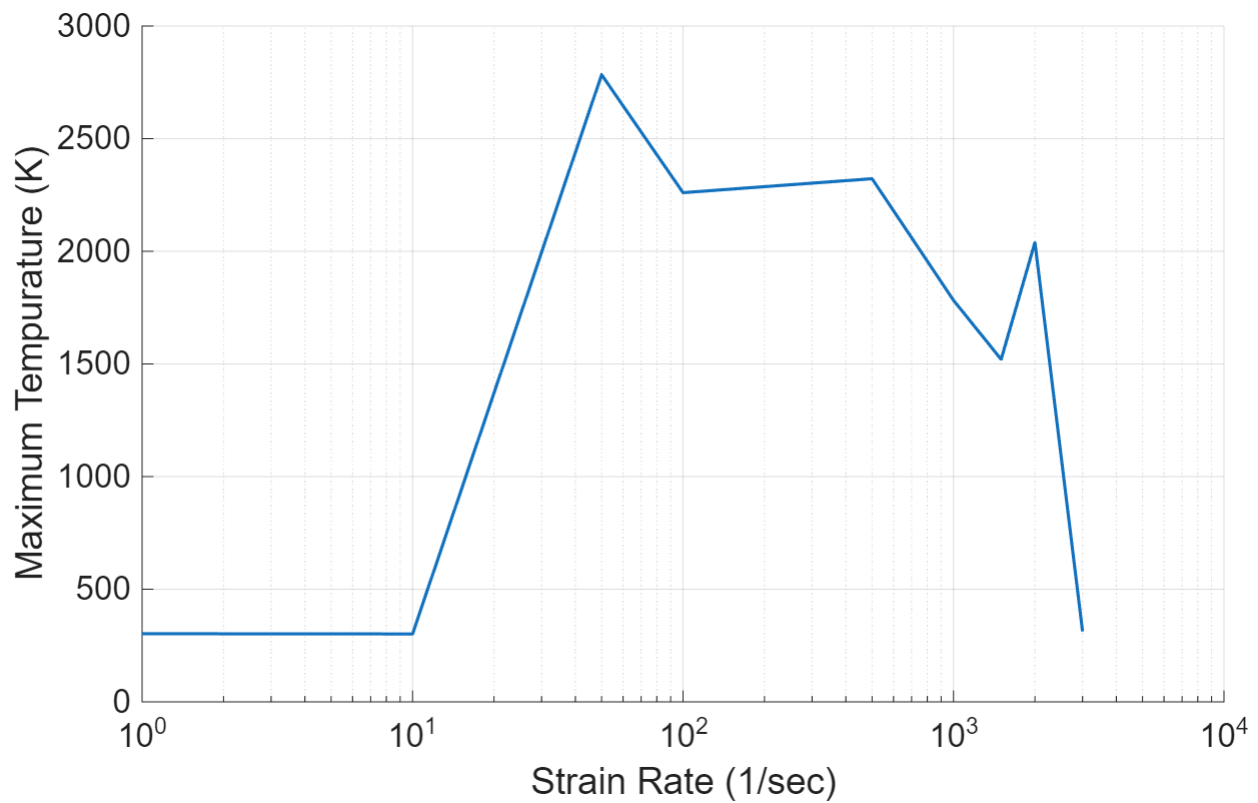


Figure 12 Maximum Temperature versus Strain Rate

Fig. 12 shows the maximum temperature at various strain rates. The maximum temperatures generally increases then decreases with increasing strain rates with the peak temperature being 2700K at a strain rate of around 50 1/sec. There is another peak in temperature at a strain rate of 1500 1/sec. The difference in maximum temperature between 1 and 10 is very small as well.

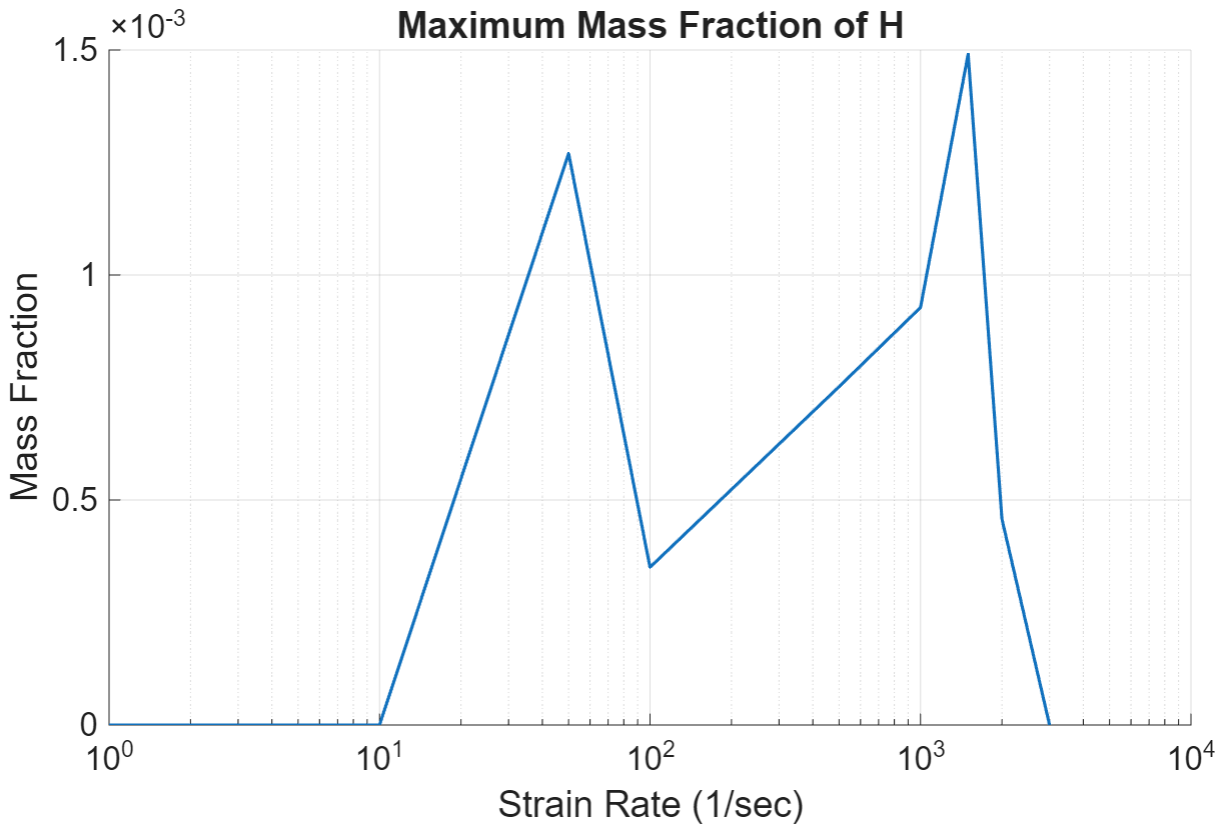


Figure 13 Maximum Mass Fraction of H versus Strain Rate

Fig. 13 shows the maximum mass fraction of H with a changing strain rate. The mass fraction has spikes at a strain rate of 50 and 1500 1/sec. These peak mass fractions are not very large, as the largest mass fraction is 0.0015 at a strain rate of 1500 1/sec. The mass fraction also does not change much from a strain rate of 1 to a strain rate of 10.

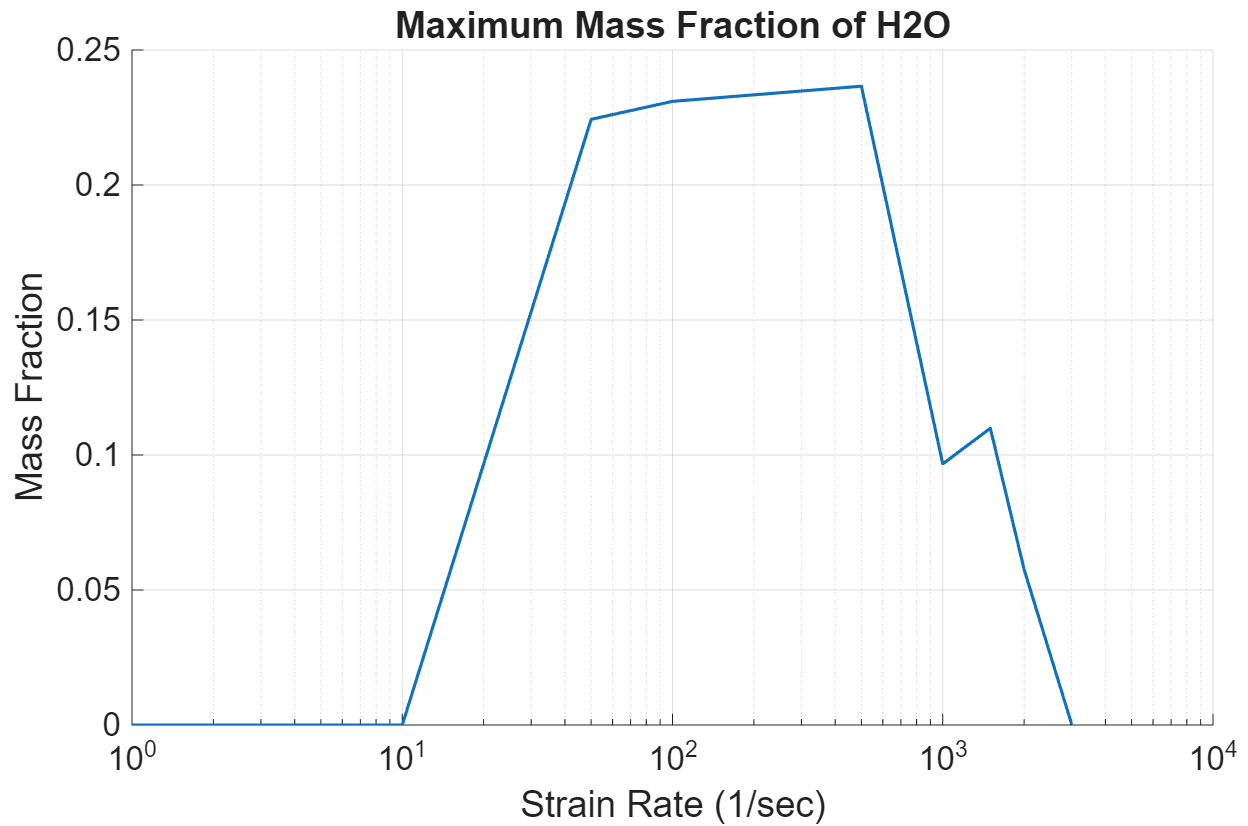


Figure 14 Maximum Mass Fraction of H₂O versus Strain Rate

Fig. 14 shows the maximum mass fraction of H₂O at different strain rates. Overall, the mass fraction increases, plateaus, then decreases. This plateau is between a strain rate of 50 and 500 1/sec with the maximum value being 0.24 which is significantly larger than the maximum mass fraction of H. There is also another very small spike at a strain rate of 1500 1/sec.